

A System to Visualize Differences in Paddling Timing between Teammates in Rowing

Daiki Takahashi
daiki.takahashi@iis.ise.ritsumei.ac.jp
Ritsumeikan University
Osaka, Japan

Kazuya Murao
murao@fc.ritsumei.ac.jp
Ritsumeikan University
Osaka, Japan

ABSTRACT

The use of wearable devices and motion sensors has had a significant impact on sports. In this study, we propose a method to improve rowing performance by analyzing 3-axis acceleration data from smartwatches worn by rowers. The goal is to enhance synchronization and timing among multiple rowers on the same boat, which is crucial for competitive rowing. We conducted experiments using a Concept2 rowing ergometer and collected data from rowers wearing Apple Watches. The collected data was processed through filtering, peak detection, and distance calculations (DTW and Euclidean). Our evaluation demonstrated that the proposed method effectively identified discrepancies, providing a quantitative basis for performance improvement. The results emphasize the potential of wearable technology in providing precise, real-time feedback to rowers, ultimately assisting in better coordination and technique refinement.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

smart watch, wearable computing, accelerometer, rowing

ACM Reference Format:

Daiki Takahashi and Kazuya Murao. 2024. A System to Visualize Differences in Paddling Timing between Teammates in Rowing. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678441>

1 INTRODUCTION

Recently, various technologies, such as wearable devices and motion sensors, which play an essential role in the sports industry, have rapidly advanced. These technological advancements have been included in rowing, where accelerometers, gyroscopes, and GPS sensors are integrated into boats to provide detailed measurements of boat movements[10]. The application of these technologies is

anticipated to enhance the performance of rowers, which directly influences the outcome of the competition.

Providing feedback on the quality of techniques and methods for improvement is a fundamental educational mechanism. Conveying accurate technical information effectively can significantly aid rowers' progress. Rowers often find it challenging to recognize their incorrect movements, which can lead to injuries. Technologies that report rowers' technical abilities in real-time can assist in coaching, offering timely and consistent objective support, mainly when coaches are unavailable.

The rowing movements comprise the catch, drive, finish, and recovery phases. Efficient repetition of these phases maximizes the propulsion of the boat. Especially in competitive rowing, coordination among multiple rowers is essential, and synchronizing the power and timing of each rower's stroke is a critical factor in determining the outcome of a race. Currently, timing synchronization primarily relies on the coxswain's calls, making speed control very challenging due to the continuous coordination required between 2 and 8 rowers and the coxswain[4]. Thus, developing feedback methods for rowers' timing is necessary.

Traditional coaching methods involve the coxswain riding at the stern and the coach observing from a motorboat, providing advice based on experiential data. Observation with the naked eye limits feedback to fundamental data such as boat speed and stroke rate for each rower, constraining quantitative analysis. Therefore, introducing technologies capable of more precise data collection and analysis is essential. There is a strong demand for computer-assisted approaches to enhance the training process in rowing[7].

Currently, sensing technologies in competitive rowing include systems for measuring rowing timing using pressure sensors and evaluating rowing movements using wearable devices[5]. However, adequate research on timing measurement methods using wearable devices is still lacking.

This paper proposes a method for acquiring 3-axis acceleration data using smartwatches worn on the wrists of each rower in scenarios where multiple rowers are rowing simultaneously. By comparing and verifying the rowing movements with the acquired 3-axis acceleration data, the aim is to construct a training environment based on quantitative data. This approach aims to provide more effective training methods and enhance rowers' performance.

Section 2 introduces related works, section 3 describes the details of the proposed method, and section 4 presents our evaluation. We conclude in section 5 with a brief summary.

2 RELATED WORK

This section introduces research on action recognition and research on rowing.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3678441>

2.1 Research on action recognition

Jenny et al.[11] investigated the accuracy of template matching for classifying sports activities using acceleration signals recorded by wearable devices. Using a wrist-worn accelerometer, they measured and classified eight common sports activities, achieving an accuracy of approximately 85%. Akin et al.[2] researched trends and applications of activity recognition using motion sensors. Miikka et al.[6] investigated how well everyday and sports activities performed by subjects could be recognized in an unsupervised environment compared to a supervised one. Activity data were collected from 12 subjects for 68 hours, of which 21 hours were supervised, and 47 were unsupervised. The overall accuracy of activity recognition using both supervised and unsupervised data was 89%, just 1% lower than the accuracy of activity recognition using only supervised data. However, when supervised data were used only for training and unsupervised data were used only for validation, the accuracy decreased by 17%. These studies suggest that wearable devices in rowing competitions allow for the validation of movements.

2.2 Research on rowing

Marco et al. [3] proposed a system to evaluate rowing performance and support training using wireless sensors. In experiments with four rowers, the system's evaluation of all performance variations matched that of a professional coach. Jordi et al. [10] proposed a system to evaluate rowing performance using a 3-axis MEMS accelerometer to track the movements of the oars and the boat. This system is stated to have better real-time performance than the existing RowX. Ross et al. [1] investigated whether real-time visual ipsative feedback could improve overall kinematic consistency in rowing. Detailed information feedback significantly improved kinematic consistency compared to no-feedback and summary-feedback interventions. Bosch et al. [5] analyzed the movements of rowers on a rowing machine using motion sensors. This research provided a preliminary analysis of the differences between experienced and novice rowers and between good and poor techniques. The study indicated that the differences between novices and experienced rowers become more pronounced as the stroke rate increases. Simon et al. [8] developed an application to provide rowers on ergometers with real-time visual feedback on their technique. The evaluation showed that this system consistently served as an effective training tool, maintaining a superior method compared to coach-led training. Hohmuth et al. [9] developed a wireless rowing measurement system (WiRMS) using an accelerometer and EMG sensors to measure boat movements and muscle activity to improve rowers' techniques. Performance evaluation using WiRMS confirmed a significant correlation ($p < 0.001$) between stroke rate and drive-recovery ratio.

These researchers have explored using various sensors in training for rowing competitions. Then, we believe that more efficient training can be provided by recognizing the timing of rowing movements among multiple rowers.

3 PROPOSED METHOD

This section describes the proposed method for collecting data using an accelerometer when rowers perform rowing movements on an ergometer and for recognizing timing discrepancies among

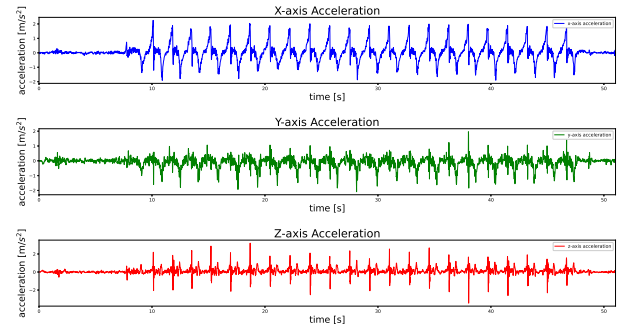


Figure 1: 3-axis acceleration of the rowing movement.

multiple rowers. This method aims to enhance athletic performance by improving the synchronization of movements.

3.1 Overview

The proposed method involves composite acceleration data from multiple rowers using Apple Watches while they perform rowing movements, and the data is compared to identify discrepancies in their movements. Specifically, the method compares the timings of the critical phases of the rowing movement, namely the catch and the finish. This approach allows for precise identification of rower timing differences, which can then be corrected. The following sections describe the data collection, filtering, peak detection, DTW distance calculation, and Euclidean distance calculation.

3.2 Data collection

In the data collection, rowers wear Apple Watches on their left wrists and perform rowing movements on an ergometer. During this process, 3-axis acceleration data are collected at a sampling frequency of 100 Hz. Rowers commonly wear the Apple Watch used in this method during regular training sessions, which does not interfere with their training. An example of the collected 3-axis acceleration data is shown in Figure 1.

3.3 Filtering

In the filtering, the composite acceleration for each athlete is calculated from the 3-axis acceleration data collected in the data collection component. By using the composite acceleration, the overall intensity of the motion can be evaluated with a single value, allowing for a comprehensive understanding of the movement's energy and intensity. Additionally, this approach saves computational resources and time, contributing to the design of future systems. The following formula gives the calculation of the composite acceleration:

$$A_t = \sqrt{x_t^2 + y_t^2 + z_t^2}$$

Here, A_t is the composite acceleration at time t . Figure 2 shows the calculated composite acceleration. Following this, we smooth the calculated composite acceleration A_t using a simple moving average. The calculation algorithm is shown below.

- (1) Define the window size W .

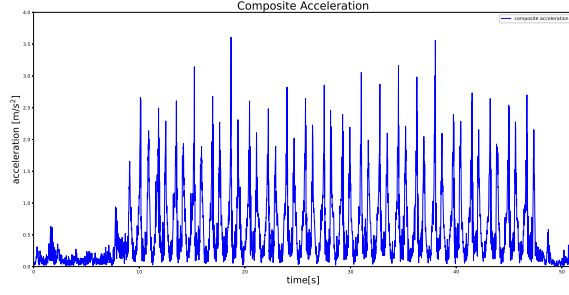


Figure 2: Composite acceleration of the rowing movements.

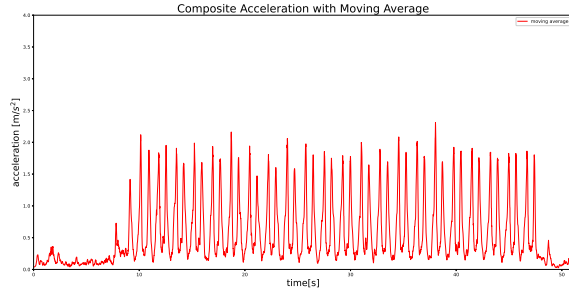


Figure 3: Composite acceleration smoothed by moving average.

- (2) Start from the first data point and segment the data into windows of size W .
- (3) Calculate the mean of the data points within each window using the following formula:

$$SMA_t = \frac{1}{W} \sum_{i=0}^{W-1} A_{t+i}$$

Here, SMA_t is the moving average at time t , and A_{t+i} represents the data point at time $t + i$.

- (4) Move the window forward by one and repeat the same calculation starting from the next data point.

The smoothed composite acceleration is shown in Figure 3.

3.4 Peak detection

In the peak detection, the primary timings of the movements (catch and finish) are identified from the filtered composite acceleration using the `find_peaks` function from the SciPy signal module. By using peaks, it becomes possible to automatically detect and segment motion cycles, enabling efficient data processing without relying on manual analysis. Additionally, this makes it easier to compare different sessions and subjects. This function detects local maxima in the signal, allowing for accurate capture of the movements cycle. The detected peaks are shown in Figure 4.

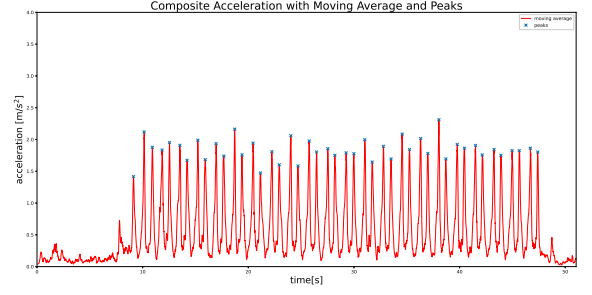


Figure 4: Detected peaks.

3.5 DTW distance calculation

The distance between the composite acceleration waveforms collected from two rowers is calculated using DTW (Dynamic Time Warping) distance. In DTW distance calculation, we compare the similarity of actions between rowers. In this process, the time series is not considered; instead, we use the peaks detected from the composite acceleration of the rowers to compare the actions segmented for each rower. The specific algorithm is shown below. When comparing two discrete waveforms of length m and n , $X = (x_1, \dots, x_m)$ and $Y = (y_1, \dots, y_n)$, we define the $m \times n$ matrix $d(i, j) = (x_i - y_j)^2$. Next, we search for the warping path $W = (w_1, \dots, w_k)$ consisting of pairs of indices from X and Y . The warping path must satisfy the following three conditions:

- Boundary condition:
 $w_1 = (1, 1), w_k = (m, n)$
- Continuity:
 $w_k = (a, b), w_{k-1} = (a', b') \Rightarrow a - a' \leq 1 \wedge b - b' \leq 1$
- Monotonicity:
 $w_k = (a, b), w_{k-1} = (a', b') \Rightarrow a - a' \geq 0 \wedge b - b' \geq 0$

To find the path with the minimum distance, we apply the following procedure:

- Initialization:
 $DTW(0, 0) = 0$
 $DTW(i, 0) = \infty$ for $i = 1, \dots, m$
 $DTW(0, j) = \infty$ for $j = 1, \dots, n$
- Do for $i = 1, 2, \dots, m$:
Do for $j = 1, 2, \dots, n$:

$$DTW(i, j) = d(x_i, y_j) + \begin{cases} DTW(i-1, j-1) \\ DTW(i-1, j) \\ DTW(i, j-1) \end{cases}$$

- Output:
Return $DTW(m, n)$

The cost $DTW(m, n)$ obtained by the above equation is the DTW distance between X and Y . The DTW distance between movements is used not to evaluate the timing of movements among rowers but to assess the movements' similarity.

3.6 Euclidean distance calculation

In calculating the Euclidean distance, we compare the composite acceleration between rowers while considering the time series. By doing this, even if similar actions are performed, if the timing is different, the distance will be greater, thus allowing us to compare the timing. The specific algorithm is outlined below.

- (1) Prepare two n -dimensional vectors $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n)$.
- (2) Calculate the Euclidean distance.

$$h(X, Y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

The cost obtained by the above formula $h(X, Y)$ represents the Euclidean distance between X and Y . The Euclidean distance is used to evaluate the timing of movements between rowers.

Peak detection, DTW distance calculation, and Euclidean distance calculation are performed on composite accelerations collected from multiple rowing movements to compare rowing movements between rowers. This allows for evaluating movements and its synchronization, enabling specific guidance.

4 EVALUATION

To evaluate the effectiveness of the proposed method, this section describes an experiment in which the proposed method was used on rowers.

4.1 Experiment environment

Two male rowers in their 20s, α and β , both with rowing experience, used the Concept2 rowing ergometer, set the distance to 300 meters on the PM5 performance monitor, and performed rowing movements. During this, they wore Apple Watches on their left wrists to collect 3-axis acceleration data. Additionally, the rowing movements were recorded on video. The experimental process is shown in Figure 5. Subsequently, three subjects in their 20s, both male and female, watched the recorded rowing video and provided feedback. Subject γ was a male with rowing experience, subject δ was a female with rowing experience, and subject ϵ was a male without rowing experience. The differences in movements between the two rowers were examined based on the 3-axis acceleration data collected by the Apple Watches and the feedback from the recorded rowing video. Movement A includes synchronized movements, while Movement B includes unsynchronized movements.

4.2 Results and discussion

Table 1 and 2 show the peak error, DTW distance, and Euclidean distance between the movements of two rowers calculated from the 3-axis accelerations collected when they simultaneously performed rowing movements. A-1 captures the movements from finish to end of recovery for the two rowers, while A-2 captures the movements from catch to end of drive. Similarly, odd-numbered movements indicate movements from finish to end of recovery, whereas even-numbered movements indicate movements from catch to end of drive.

Table 3 lists the movements where the two rowers were perceived to be unsynchronized by three subjects who provided feedback



Figure 5: Data collection on the ergometer

Table 1: Comparison results for movement A.

No.	Peak error [ms]	DTW dist.	Euclidean dist.
A-1	60	11.834	1.466
A-2	120	7.908	2.398
A-3	30	14.400	4.623
A-4	90	10.137	1.131
A-5	70	13.657	2.152
A-6	120	9.251	2.290
A-7	90	12.399	4.091
A-8	160	9.888	3.415
A-9	90	11.691	4.290
A-10	150	10.948	3.243
A-11	110	10.963	5.029
A-12	170	14.646	4.476
A-13	110	11.944	6.290
A-14	160	9.663	4.609
A-15	120	10.358	4.882
A-16	210	12.074	3.954
A-17	130	12.803	4.911
A-18	180	9.035	5.427
A-19	140	12.567	5.437
A-20	190	10.057	5.669

using the recorded rowing video. Based on the results in Table 3, while Movement A-2 was recognized as erroneous by all subjects, no significant differences were observed in peak error or Euclidean distance as shown in Table 1. Examples of synchronized movements' waveforms are shown in Figure 7.

The reason for this could be that in Movement A-2, rower β was seen twisting their wrists at the end of the drive, and this movement was misinterpreted as an error in the subjects' feedback. Therefore, although no significant differences were observed in comparison with acceleration, it is worth considering whether such movements can be recognized in the future by combining angular velocity and other factors.

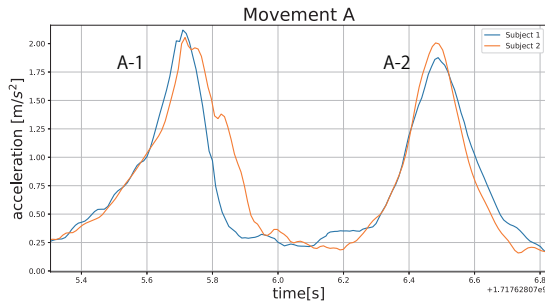
Based on the results in Table 3, in Movement B, feedback from subjects with rowing experience identified errors in 18 or 19 out of 20 movements, while novices identified errors in 16 out of 20

Table 2: Comparison results for movement B.

No.	Peak error [ms]	DTW dist.	Euclidean dist.
B-1	290	15.563	8.666
B-2	220	14.582	8.824
B-3	330	12.575	8.867
B-4	250	12.492	8.309
B-5	370	10.432	9.300
B-6	310	10.361	9.329
B-7	450	13.308	10.798
B-8	370	14.436	10.860
B-9	350	17.563	9.570
B-10	270	13.194	8.634
B-11	320	13.672	10.193
B-12	260	13.351	8.792
B-13	420	13.933	9.885
B-14	330	10.918	9.588
B-15	330	12.560	9.832
B-16	220	10.409	8.097
B-17	420	12.462	9.881
B-18	330	11.151	8.807
B-19	440	12.293	9.661
B-20	360	12.874	9.713

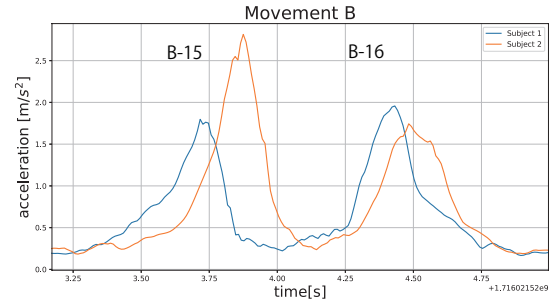
Table 3: List of recognized errors.

Movement	Subject		
	γ	δ	ϵ
A	2	2	2
B	Except 5	Except 3, 19	Except 15, 16, 18, 19

**Figure 6: Examples of synchronized movement.**

movements. Examples of unsynchronized movements' waveforms are shown in Figure 7.

Based on the results in Table 2, a common factor in movements where experienced subjects did not identify errors in Movement B was the proximity of DTW distance. This suggests that they appear similar when the two movements are not considered in terms of time series. Therefore, while timing errors may exist to some extent, there is a tendency for inherently similar movements not to be recognized as errors. Additionally, for experienced rowers, all the asynchronous movements that were overlooked occurred at

**Figure 7: Examples of unsynchronized movement.**

the finish, suggesting that providing feedback at the finish is more complicated than at the catch.

5 CONCLUSION

In this paper, we propose a method to recognize deviations in rowing movements using an accelerometer. The proposed method involves attaching Apple Watches to the rower's left wrists to acquire composite acceleration data of the rowing movements. To evaluate the effectiveness of the proposed method, we conducted an evaluation experiment between two rowers. Our future plans include developing a system for real-time feedback. Additionally, considering actual training scenarios, we aim to improve the method to recognize errors in parts of the stroke other than the finish and catch.

REFERENCES

- [1] Ross Anderson, Andrew Harrison, and Gerard M Lyons. 2005. Rowing: Accelerometry-based feedback-can it improve movement consistency and performance in Rowing? *Sports biomechanics* 4, 2 (2005), 179–195.
- [2] Akin Avci, Stephan Bosch, Mihai Marin-Perianu, Raluca Marin-Perianu, and Paul Havinga. 2010. Activity recognition using inertial sensing for healthcare, wellbeing and sports applications: A survey. In *23th International conference on architecture of computing systems 2010*. VDE, 1–10.
- [3] Marco Avvenuti, Daniel Cesarini, and Mario GCA Cimino. 2013. MARS, a multi-agent system for assessing rowers' coordination via motion-based stigmergy. *Sensors* 13, 9 (2013), 12218–12243.
- [4] Alexandre Baudouin and David Hawkins. 2004. Investigation of biomechanical factors affecting rowing performance. *Journal of biomechanics* 37, 7 (2004), 969–976.
- [5] Stephan Bosch, Muhammad Shoaib, Stephen Geerlings, Lennart Buit, Nirvana Meratnia, and Paul Havinga. 2015. Analysis of indoor rowing motion using wearable inertial sensors. In *Proceedings of the 10th EAI international conference on body area networks*. 233–239.
- [6] Miikka Ermes, Juha Pärkkä, Jani Mäntytjärvi, and Ilkka Korhonen. 2008. Detection of daily activities and sports with wearable sensors in controlled and uncontrolled conditions. *IEEE transactions on information technology in biomedicine* 12, 1 (2008), 20–26.
- [7] Chien-Liang Fok, Dushyanth Balasubramanian, Mark Tamola, and Chenyang Lu. 2004. *TinyCoxswain: Using a Sensor Network to Enhance Crew Performance*. Technical Report. Citeseer.
- [8] Simon Fothergill. 2010. Examining the effect of real-time visual feedback on the quality of rowing technique. *Procedia Engineering* 2, 2 (2010), 3083–3088.
- [9] Richard Hohmuth, Daniel Schwensow, Hagen Malberg, and Martin Schmidt. 2023. A wireless rowing measurement system for improving the rowing performance of athletes. *Sensors* 23, 3 (2023), 1060.
- [10] Jordi Llosa, Ignasi Vilajosana, Xavier Vilajosana, Nacho Navarro, Emma Surinach, and Joan Manuel Marques. 2009. REMOTE, a wireless sensor network based system to monitor rowing performance. *Sensors* 9, 9 (2009), 7069–7082.
- [11] Jenny Margarito, Rim Heloui, Anna M Bianchi, Francesco Sartor, and Alberto G Bonomi. 2015. User-independent recognition of sports activities from a single wrist-worn accelerometer: A template-matching-based approach. *IEEE Transactions on Biomedical Engineering* 63, 4 (2015), 788–796.